Numerical solution of ODE by Euler's method.

In Euler's method, if the differential equation is

$$\frac{dy}{dx} = f(x, y)$$

then, the general solution is given by,

$$y_{n+1} = y_n + h \cdot f(x_n, y_n)$$

or

$$y_n = y_{n-1} + h \cdot f(x_{n-1}, y_{n-1})$$

① Find y_1, if $\frac{dy}{dx} = \log(x+y)$

$y(0) = 2$ at $x = 0.2$ taking $h = 0.2$

$(0, 0.2)$ → Interval

$n = 1$

solution,

there, $x_0 = 0$ & $y_0 = 2$

y_1 at $x = 0.2$, $h = 0.2$

$$f(x, y) = \log(x+y)$$

$$f(x_0, y_0) = \log(0+2)$$

$$= \log(2)$$

$$= 0.0310$$

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Now, In general

$$y_n = y_{n-1} + h \cdot f(x_{n-1}, y_{n-1})$$

$$y_1 = y_0 + h \cdot f(x_0, y_0)$$

$$y_1 = 2 + 0.2 \times 0.0310$$

$$= 2.0602$$

(2) use Euler's method to find an approximation value of y correct upto 4 decimal places for $x = 0.1$ given $\frac{dy}{dx} = x - y^2$ at $x = 0$, $y = 1$ taking $h = 0.02$.

Solution:-

$a = 0$ & $b = 0.1$

$$n = \frac{b-a}{h}$$

$$= \frac{0.1 - 0}{0.02} \times \frac{100}{100}$$

$$= 5$$

$x_0 = 0$

$y_0 = 1$

$f(x, y) = x - y^2$

To Find y at $x = 0.1$

$(0, 0.1)$ into 5 steps

$$x_n = x_{n-1} + h$$

$$x_1 = x_0 + 0.02$$

$$= 0.02$$

$$x_2 = x_1 + h$$

$$= 0.04$$

$(0, 0.02, 0.04, 0.06, 0.08, 0.1)$

$$F(x, y) = x - y^2$$

$$F(x_0, y_0) = x_0 - y_0^2 = 0 - 1 = 1$$

~~$F(x_1, y_1) =$~~

using Euler's method

$$y_n = y_{n-1} + h \cdot F(x_{n-1}, y_{n-1})$$

$$y_1 = y_0 + h \cdot F(x_0, y_0)$$

$$y_1 = 1 + (0.02) \times (-1)$$

$$= 0.98$$

after find

$$F(x_1, y_1) = x_1 - y_1^2 = 0.02 - 0.98^2$$

$$= -0.9404.$$

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$$y_2 = y_1 + h \times F(x_1, y_1)$$

$$= 0.98 + (0.02 \times -0.98404)$$

$$= 0.9612$$

$$F(x_2, y_2) = x_2 - y_2^2$$

$$= -0.8839$$

$$y_3 = y_2 + h \times F(x_2, y_2)$$

$$= 0.9434.$$

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